# Practice problems #15

1. Let $A = \begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$.

   (a) Find an $LU$ factorization of $A$.

   (b) Use (a) to solve the equation $Ax = b$.

2. True or False. Justify your answer:
   Suppose a matrix $A$ is row reduced to an echelon form $U$. If $A = LU$ is an $LU$ factorization, then $A$ can be row reduced to $U$ using only replacement operations.

   (This is asking the *converse* of what we did in class. In class, we showed that *if* $A$ is row reduced to $U$ using only replacement operations, *then* $A = LU$.)

Answers or Hints on page 2.

#1: $L = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -5 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{bmatrix}$, $y = \begin{bmatrix} 0 \\ -5 \\ -18 \end{bmatrix}$, $x = \begin{bmatrix} -5 \\ 1 \\ 3 \end{bmatrix}$

#2: True. (Since $L$ means a unit lower triangular matrix in the $LU$ factorization, $L$ can be reduced to $I$ using only which row operation? So, $E_k \cdots E_1 L = I$ where $E_i$ are what kind of elementary matrices? Then, $A = LU$ becomes $E_k \cdots E_1 A = E_k \cdots E_1 LU$.)